

A Multimodal Methodology for Cricket Video Event-Player Association and Retrieval

Abstract

Cricket match video analysis is important for performance evaluation, coaching, tactical review, content retrieval, and viewer engagement. Nevertheless, locating player-specific actions and key match events in full-length cricket recordings usually requires manual tagging or extensive video review. This research proposes a multimodal methodology for cricket video event-player association and retrieval using scoreboard OCR, umpire-signal recognition, named entity recognition (NER), and timestamp synchronization. The proposed framework extracts scoreboard information using EasyOCR, identifies player entities using NER and dictionary-based matching, detects key umpire-signal events using EfficientNet-B0 and ResNet-18, and associates detected events with player roles through timestamp alignment. The generated event-player records are organized into a structured retrieval layer supporting player-wise filtering, event-based navigation, wicket attribution, and cricket-domain query retrieval. Experimental results show that umpire detection achieved 0.91 accuracy and 0.92 weighted F1-score, while event classification achieved 0.83 accuracy and 0.83 macro F1-score. Scoreboard OCR achieved a normalized exact-match accuracy of 0.9211, and the full event-player association rate reached 0.741. The NER-supported player identification approach improved overall role accuracy from 0.79 to 0.83. The structured retrieval mechanism correctly answered 190 out of 200 cricket-domain queries, achieving 95.0% retrieval accuracy. The proposed methodology reduces manual effort by generating structured, timestamped cricket video segments that can be searched and retrieved using player identity and event type.

Keywords: Cricket video analysis, player identification, video segmentation, named entity recognition, umpire signal recognition, sports video analytics

1 Introduction

Cricket produces large volumes of broadcast video content through international tournaments, domestic leagues, franchise competitions, and televised match recordings. Since cricket has a large global viewership and matches can last for several hours or even days, broadcast cricket videos are naturally lengthy and require efficient methods for detecting and retrieving important events [1, 2]. These videos contain valuable information for performance analysis, tactical decision-making, coaching, post-match

review, fan engagement, and digital content creation. A single match can contain hundreds of deliveries, multiple player changes, several scoreboard updates, boundaries, wickets, appeals, and umpire signals. Although all of these moments are useful, they are usually embedded inside long video recordings. As a result, locating a particular player’s event or a specific match moment often requires manual video review.

Traditionally, coaches, analysts, and viewers depend on manually prepared highlights, human tagging, or repeated video navigation to locate boundaries, wickets, dismissals, or player-specific performances [1–4]. This manual process is slow and inconsistent because it depends on the availability of an analyst and the quality of manual annotations. For example, if a coach wants to review all boundaries scored by a particular batsman, or if an analyst wants to identify wickets taken by a selected bowler, the relevant moments must be found by scanning through the video or by relying on pre-existing highlights. Nevertheless, highlight videos often contain only selected moments and may not preserve the full player-event context. Consequently, a structured retrieval framework that connects video segments with player identity, event type, and timestamp is useful.

Automated sports video understanding has become an important research area. Computer vision and natural language processing techniques have been used for event detection, player recognition, action recognition, scene classification, highlight generation, video summarization, and content-based retrieval [1–3, 5–9]. In cricket, however, the structure of the game creates additional challenges. Important information is distributed across multiple sources such as scoreboard overlays, player names, score changes, umpire gestures, visible event signals, and timestamps. Some events can be inferred visually through umpire signals, while player names are often available only through the scoreboard. Consequently, relying on one source of information is not enough for player-specific retrieval.

Many existing approaches focus mainly on event detection, scene classification, wicket segmentation, summarization, ball-delivery segmentation, or player-specific highlight generation [1, 2, 6, 7, 10–19]. These methods can identify important moments, but they often do not connect those events with the responsible batsman, bowler, and exact timestamp. Similarly, OCR-based frameworks can read scoreboard text, but OCR alone cannot reliably identify visual event types such as FOUR, SIX, or WICKET. This creates a gap between detecting an event and making that event searchable by player identity. For practical use, a cricket video retrieval framework should answer questions such as “show the first six”, “show wickets”, “show Kohli batting”, “show Kohli out”, or “show wickets by Bumrah”. To answer such queries, the framework must connect event type, player identity, role information, and video timestamp.

The key research problem addressed in this paper is the absence of an integrated cricket-specific framework that jointly performs event detection, player identification, event-player association, timestamp synchronization, video segmentation, and dashboard-based retrieval. Existing event detection methods may identify major moments using visual cues, scoreboard context, or umpire gestures [20–25]. However, they generally do not produce structured player-event records. Existing summarization

approaches may generate shorter videos, but they do not always support user-controlled retrieval by player and event type. Consequently, this research focuses on a practical end-to-end pipeline that converts cricket broadcast videos into structured, searchable, timestamped event records.

To address this gap, this research proposes an automated cricket video segmentation and retrieval framework. Scoreboard information is extracted using EasyOCR [13]. Key cricket events are detected through umpire-signal recognition using EfficientNet-B0 for umpire detection [26] and ResNet-18 for pose classification [11]. Player entities are identified using NER and dictionary-based player-name matching [12]. Detected events and player information are synchronized using timestamps and stored as structured event records. These records are then presented through a structured retrieval layer that supports player-based filtering, event navigation, wicket attribution, and query-based retrieval.

The proposed approach does not attempt to solve every possible cricket video understanding problem. Instead, it focuses on major implemented event categories and retrieval tasks supported by the developed pipeline. The event classifier focuses on FOUR, SIX, OUT, and NO-ACTION classes. The structured retrieval is rule-based and operates over generated event-player records. The framework is designed to reduce manual video review by automatically generating searchable video segments from the available visual and scoreboard information [9, 18, 22].

The main contributions of this research are:

- An end-to-end multimodal cricket video retrieval framework that integrates scoreboard OCR, umpire-signal event detection, NER-supported player identification, and timestamp-based event-player association.
- A two-stage cricket event recognition approach that first detects umpire presence using EfficientNet-B0 and then classifies umpire gestures using ResNet-18 to reduce false event predictions.
- A player-event association method that links detected cricket events with batsman, bowler, and timestamp information extracted from scoreboard context.
- A structured video segmentation workflow that generates playable video segments around detected events and stores event-player metadata for retrieval.
- A rule-based structured retrieval layer that enables player-wise, event-wise, ordinal, dismissal, and bowler-wicket video navigation from structured event records.
- A component-level and framework-level evaluation covering umpire detection, event classification, OCR accuracy, player identification, event-player association, query retrieval, and baseline comparison.

The remainder of this paper is organized as follows. Section 2 reviews related work in cricket video analysis, sports video summarization, multimodal event detection, and retrieval. Section 3 explains the proposed methodology, including dataset preparation, preprocessing, OCR, event detection, player identification, event-player association, segmentation, and structured retrieval. Section 4 presents experimental results and discussion. Section 5 concludes the paper and outlines future improvements.

2 Literature Review

Sports video analysis has been widely studied for event detection, player tracking, highlight generation, action recognition, scene classification, semantic event classification, and content-based video retrieval. Earlier studies explored visual, audio, and contextual cues for identifying important events in sports videos, while recent studies have applied deep learning models to improve scene, action, and event classification accuracy [1, 3, 6–8]. Cricket video analysis has also received attention because cricket matches contain long video durations, frequent scoreboard updates, umpire gestures, player-specific actions, and event-based moments such as boundaries and wickets [1–3, 5, 7, 10].

2.1 Cricket Event Detection and Summarization

Raval and Goyani [1] reviewed cricket event detection and video summarization approaches and highlighted the importance of automated methods for reducing manual video analysis. Their review shows that cricket video summarization can help users access meaningful match content without watching full-length videos. Nevertheless, their work mainly focused on summarization techniques and did not provide an implemented end-to-end framework for linking detected events with specific batsmen, bowlers, and timestamps.

Shingrakhia and Patel [2] reviewed cricket highlight generation methods and discussed how important match moments can be extracted from video. Highlight generation is useful because it reduces the length of video content and presents key moments to viewers. Nevertheless, highlight-focused frameworks usually provide a summarized output rather than a searchable player-event index. This limits their usefulness for tasks such as retrieving all events associated with a particular player or locating the exact dismissal of a batsman.

Nasir et al. [3] proposed cricket event detection and summarization using multi-modal information. Their work showed that cricket events can be detected using a combination of video-related cues. This is relevant to the present research because cricket events are often represented by more than one type of information. Nevertheless, their method did not provide detailed player-event association. As a result, detected events were not directly linked with the responsible batsman, bowler, or player role.

Kolekar et al. [5] investigated semantic event detection and classification in cricket video sequences. Their approach contributed to cricket event labelling and showed the importance of semantic understanding in sports video analysis. Nevertheless, it did not integrate scoreboard OCR, player-name extraction, or dashboard-based retrieval for user queries. Consequently, although event detection was addressed, player-wise retrieval remained limited.

2.2 Umpire-Signal Recognition and Scoreboard Automation

Umpire-signal recognition is closely related to cricket event detection because official signals visually confirm events such as boundaries, wickets, wides, and no-balls.

Earlier work by Hari and Wilscy [23] detected cricket events using intensity projection profiles of umpire gestures and machine learning-based classification. Ravi et al. [24] introduced the SNOW dataset for umpire pose detection and used deep features extracted from pretrained CNN models with SVM classification to recognize umpire signals such as SIX, NO BALL, OUT, and WIDE. These studies show that umpire gestures can be used as meaningful visual cues for cricket event recognition, but they mainly focus on gesture/event classification rather than player-specific video retrieval.

Sakib et al. [20] proposed an event detection approach using video-based CNN classification of umpire signals. Their method used video-based I3D modelling for umpire-signal recognition and introduced a dataset containing 504 videos and 2000 images from cricket matches. Their work detected key events such as FOUR, SIX, OUT, NO BALL, and WIDE from official cricket match videos. This work is directly relevant to the present research because both approaches use umpire signals as visual evidence for cricket event detection. Nevertheless, their work mainly focused on event detection from umpire signals and did not provide player-identity association, batsman and bowler role extraction, timestamp-based event-player linking, or dashboard-based retrieval.

Dey et al. [21] proposed an attention-based DC-GRU network for cricket umpire signal recognition. Their work introduced a deep temporal modelling approach for recognizing umpire actions and showed the importance of learning signal patterns from cricket videos. More recently, Mehta and Sharma [25] proposed a hybrid CNN-SVM framework for automated detection and classification of cricket umpire actions. These studies are relevant because umpire gestures are also used as the main visual cue for event recognition in the present work. Nevertheless, the present research differs by integrating umpire-signal classification with scoreboard OCR, NER-supported player identification, event-player timestamp synchronization, and query-based retrieval.

Nair et al. [22] studied cricket scoreboard automation using umpire gesture recognition. This indicates that umpire signals can be used to reduce manual scoring effort and automatically update cricket scores. Nevertheless, scoreboard automation and video retrieval are different objectives. Scoreboard automation focuses on updating match score values, while the present study focuses on generating structured video records that connect event type, batsman, bowler, and timestamp for later retrieval. Consequently, existing umpire-signal recognition and scoreboard automation studies support the importance of visual event recognition in cricket. Nevertheless, they generally do not address the complete retrieval problem where detected events must be linked with player identity and made searchable through a dashboard.

2.3 Multimodal Sports Video Analysis

Javed et al. [6] proposed a multimodal cricket video summarization framework using audio-visual features. This study demonstrated the usefulness of combining multiple modalities for cricket video understanding. Multimodal processing is important because cricket events may not always be clear from a single source. For example, the scoreboard may show player context, while the umpire signal may confirm the event type. Nevertheless, the output of this work was mainly a summarized video representation rather than a structured player-event index. Consequently, it did not allow

users to directly retrieve events such as a particular batsman’s boundaries, a bowler’s wickets, or a specific dismissal timestamp.

Bhalla et al. [27] proposed a multimodal approach for automatic cricket video summarization using OCR, sound detection, and replay detection. Their work is relevant because it shows that cricket event summarization benefits from combining multiple information sources rather than relying on a single cue. Similarly, Saha et al. [18] proposed a cricket highlight generation framework using score extraction and action recognition. These studies support the need for multimodal cricket video processing. Nevertheless, their outputs are mainly highlight summaries, while the proposed approach focuses on timestamped player-event retrieval.

Tang et al. [7] studied highlight detection in sports videos using cricket as a test case. Their approach supported automatic highlight detection and contributed to sports video retrieval research. Nevertheless, the method did not focus on cricket-specific player identity extraction or event-player timestamp synchronization. This is an important limitation for player-specific retrieval because a highlight is more useful when it is linked to the relevant player and event context.

Kolekar and Sengupta [8] also studied semantic concept mining in cricket videos for automated highlight generation. Their work is relevant because it attempts to move beyond low-level visual features toward meaningful cricket concepts. Nevertheless, the present research differs by focusing on a practical retrieval framework that associates detected events with scoreboard-derived player roles and makes the results available through a dashboard.

Recent work has further strengthened the role of deep learning and multimodal processing in sports and cricket video analysis. Rafiq et al. [28] used transfer learning for cricket scene classification and classified scenes such as batting, bowling, boundary, crowd, and close-up for sports video summarization. Sadlier and O’Connor [29] proposed an audio-visual event detection framework for broadcast field-sports videos using features from multiple modalities and support vector machine-based classification, showing the usefulness of combining visual and audio cues for sports event detection. Narwal et al. [30] proposed a dynamic and personalized cricket video summarization approach using a multi-modal multi-module architecture with umpire detection, umpire pose recognition, and scoreboard video OCR.

Xu et al. [31] reviewed recent deep learning methods for sports video event detection and highlighted temporal action localization, action spotting, and precise event spotting as important directions for temporally accurate sports analytics. Ferdous et al. [14] proposed an OCR-guided YOLOv8 framework for wicket-taking delivery segmentation and trajectory-based dismissal-zone analysis in cricket videos. These recent studies show that sports video analysis is moving toward deep learning, multimodal processing, OCR-guided event localization, and timestamp-aware video understanding. Nevertheless, they still do not fully address player-specific event retrieval by connecting event type, player role, and timestamped retrieval navigation in a single cricket-focused workflow.

Additional cricket-focused studies have also explored deep learning and multimodal processing for highlight generation, scene classification, and event summarization. Mahum et al. [15] proposed a player-specific framework for cricket highlight generation

using deep convolutional neural networks, showing the importance of linking cricket highlights with player identity. Shingrakhia and Patel [16] proposed a hybrid machine learning model that combines excitement, object, and event-based features for cricket video summarization. Jha et al. [17] used scoreboard region extraction and OCR for semi-automatic cricket video summarization. These studies confirm the value of OCR, event detection, and player-aware processing in cricket video analysis. Nevertheless, most of them focus on generating highlights or summaries rather than creating structured player-event records that support query-based retrieval by batsman, bowler, event type, and timestamp.

2.4 Need for Player-Event Association

Cricket has unique event structures compared with many other sports. Important moments such as boundaries and wickets are often confirmed through umpire signals and scoreboard updates rather than player motion alone. A boundary can be visually indicated by the umpire, while the scoreboard can identify the current batsmen and bowler. Consequently, visual event recognition and OCR-based scoreboard extraction provide complementary information.

Prior cricket video summarization and event detection studies have used visual, audio, and textual cues to identify key events [3, 5, 6, 8]. Nevertheless, scoreboard OCR can be noisy due to motion blur, low resolution, changing layouts, compressed video quality, stylized fonts, and partial occlusion. These issues make player identification difficult without additional text cleaning, entity recognition, and dictionary-based matching. Similarly, visual event recognition alone can detect a gesture but cannot identify the batsman or bowler responsible for the event. This demonstrates the need for a multimodal method.

A player-event association framework must perform several steps. First, it must detect the event type. Second, it must extract player names and roles from the scoreboard context. Third, it must align both sources using timestamps. Finally, it must store the result as a structured record that can be searched later. This differs from conventional highlight generation because the output is not only a shortened video, but a searchable index of event-player relationships.

Earlier cricket video analysis studies also explored OCR-based processing, semantic annotation, and delivery-level segmentation. Anjum et al. [4] used OCR-based score caption extraction for sports highlight generation. Sharma et al. [9] addressed fine-grain annotation of cricket videos using weakly supervised alignment between cricket videos and textual commentary. Abbas et al. [19] proposed a deep-learning-based approach for segmenting cricket ball deliveries and tracking ball movement in broadcast footage. These works are important because they show that textual cues, semantic annotation, and video segmentation can support cricket video understanding. Nevertheless, they do not fully address player-event association where visual event confirmation, noisy scoreboard handling, role extraction, timestamp synchronization, and structured retrieval must work together.

2.5 Research Gap

Existing cricket video analysis studies have made important progress in event detection, scene classification, highlight generation, wicket segmentation, semantic segmentation, OCR-based summarization, umpire-signal recognition, scoreboard automation, and multimodal cricket video summarization [1–3, 5–9, 14–25, 27]. Review-based studies such as Raval and Goyani [1] and Shingrakhia and Patel [2] show that cricket video summarization and highlight generation are important research directions for reducing manual video review. Nevertheless, these review studies mainly summarize existing approaches and do not provide an implemented player-event retrieval framework.

Several studies have focused on cricket event detection and semantic event classification. Nasir et al. [3] used multimodal information for cricket event detection and summarization, while Kolekar et al. [5] studied semantic event detection and classification in cricket video sequences. Tang et al. [7] also investigated highlight detection in sports videos using cricket as a test case. These studies show that cricket events can be identified from visual, audio, and contextual cues. Nevertheless, their outputs mainly support event detection, classification, skimming, or highlight extraction, rather than direct retrieval of player-specific moments linked with batsman and bowler roles.

Multimodal summarization studies further confirm the importance of combining different information sources in cricket video analysis. Javed et al. [6], Bhalla et al. [27], Narwal et al. [30], Shingrakhia and Patel [16], and Jha et al. [17] used combinations of audio-visual features, scoreboard OCR, replay detection, score extraction, and cricket-specific cues for summarization or highlight generation. Similarly, Rafiq et al. [28] used transfer learning for sports scene classification, while Xu et al. [31] highlighted the growing importance of temporal action localization and event spotting in sports videos. These studies are useful for identifying important match moments, but most of them produce summaries, highlights, scene labels, or event lists instead of structured player-event records that can be searched by batsman, bowler, event type, and timestamp.

Umpire-signal recognition and scoreboard automation studies are especially relevant to the present research because the proposed approach also uses umpire gestures as visual event cues. Hari and Wilscy [23], Ravi et al. [24], Sakib et al. [20], Dey et al. [21], and Mehta and Sharma [25] showed that umpire signals can be recognized using handcrafted features, deep features, and deep learning-based approaches, while Nair et al. [22] focused on scoreboard automation using umpire gestures. These studies confirm that umpire gestures are useful for identifying cricket events. Nevertheless, their main objective is event recognition or scoreboard updating, not generating searchable video segments associated with the active batsman, non-striker, bowler, and timestamp.

Other cricket-focused studies have addressed player-specific highlights, wicket delivery segmentation, OCR-based score caption extraction, fine-grain annotation, and delivery-level segmentation. Mahum et al. [15] proposed a player-specific cricket highlight generation framework, Ferdous et al. [14] focused on OCR-guided wicket-taking delivery segmentation and dismissal-zone analysis, Anjum et al. [4] used OCR-based score caption extraction for sports highlight generation, Sharma et al. [9] explored fine-grain cricket video annotation, and Abbas et al. [19] studied cricket delivery segmentation and ball tracking. These studies are related to parts of the proposed work, but they do not fully integrate umpire-signal event detection, scoreboard-based player

identification, batsman and bowler role association, timestamp synchronization, video segmentation, and dashboard-based query retrieval in one end-to-end framework.

Consequently, the research gap addressed in this study is the lack of an integrated cricket-specific video retrieval pipeline that connects event type, player identity, player role, timestamp, and playable video segment in a searchable dashboard. Existing systems mainly focus on event detection, score updating, annotation, summarization, or highlight generation. In contrast, the proposed approach is positioned as a player-event retrieval framework that supports practical navigation tasks such as retrieving a batsman’s boundaries, locating a dismissal moment, identifying wickets associated with a bowler, and accessing timestamped cricket video segments through dashboard-based queries.

3 Methodology

The proposed approach processes broadcast cricket videos to detect key events, identify players, associate players with detected events, and generate structured records for dashboard-based retrieval. The main stages are dataset preparation, video preprocessing, scoreboard OCR, umpire-signal event detection, player identification, event-player association, video segmentation, and structured retrieval, as shown in Fig. 1.

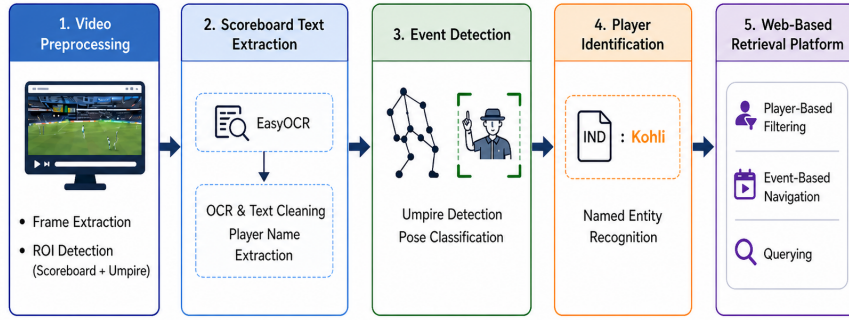


Fig. 1 High-level architecture of the proposed approach

3.1 Framework Overview

The framework follows a multimodal design because cricket video understanding requires both visual and textual information. The visual stream is used to detect umpire presence and classify umpire gestures. The textual stream is used to extract scoreboard information and identify players. These two streams are then merged

using timestamp synchronization. The final output is a structured event-player record containing the event type, timestamp, player roles, and video segment reference.

The workflow begins when a cricket video is uploaded to the processing pipeline. The video is processed frame by frame at fixed intervals. For each selected frame, the framework stores the frame index and timestamp. The lower scoreboard region is cropped and passed to EasyOCR. OCR text is cleaned and converted into candidate player tokens. At the same time, frames are passed through the umpire detection model. If an umpire is detected, the frame is passed to the pose classification model to determine the event class.

The player identification module combines OCR-based tokens, NER-based extraction, and fuzzy dictionary matching. This helps improve robustness when the OCR output contains partial names, spelling mistakes, or extra scoreboard text. The event-player association module then aligns the detected event timestamp with the closest available scoreboard/player information. Finally, the dashboard provides retrieval options for player-wise browsing, event-type filtering, wicket attribution, and cricket-domain query patterns.

3.2 Dataset Preparation and Annotation

Three datasets were prepared for the proposed approach: an umpire detection dataset, an umpire pose classification dataset, and a scoreboard OCR dataset. The datasets were created using more than 100 publicly available cricket match videos collected from YouTube. These videos represented more than 100 different matches and included footage from more than 10 teams. The selected videos covered different match styles, including international matches, domestic matches, league matches, full match recordings, and highlight clips.

The videos had mixed 720p and 1080p resolutions. This introduced variation in teams, players, camera angles, lighting conditions, scoreboard layouts, umpire positions, backgrounds, and broadcast styles. This variation was intentionally included to reduce dataset bias and improve the robustness of the proposed approach across different broadcast conditions. Since scoreboard designs are not identical across broadcasters, using different videos helped expose the OCR pipeline to multiple text layouts and visual styles.

The umpire detection dataset was manually annotated into two classes: *umpire* and *non-umpire*. The purpose of this dataset was to train a binary classifier that filters frames before pose classification. This is significant because applying pose classification to all frames can produce unnecessary false event predictions when no umpire is visible. The non-umpire class contained frames with players, field background, scoreboard content, crowd, and other non-signal visuals.

The umpire pose classification dataset is 2,000 labelled images and annotated into FOUR, SIX, OUT, and NO-ACTION classes according to the visible umpire signal. The NO-ACTION class was included because an umpire may appear in a frame without signalling a major event. Without this class, the model may incorrectly force every umpire frame into a cricket event class. The annotation process was carried out manually by the researcher and verified with the support of a cricket player to improve cricket-domain accuracy. Ambiguous, unclear, and low-quality frames were reviewed

carefully, and labels were assigned based on visible umpire gestures and cricket event context.

The umpire pose dataset was divided into 1,600 training images and 400 testing images. The scoreboard OCR dataset was prepared by extracting scoreboard regions from broadcast frames and manually verifying player-related information, including batsman 1, batsman 2, and bowler names. The umpire detection dataset was also organized into separate training and testing sets. To reduce train-test leakage, visually similar screenshots and near-duplicate frames from the same short event sequence were kept within the same split where possible. The manually verified labels were used as ground truth for evaluating OCR accuracy, player identification, and event-player association.

3.3 Video Processing and Frame Extraction

Input match videos are processed by extracting frames at fixed time intervals. Each frame is stored with its timestamp so that detected events and player information can be linked to exact video locations. Timestamp preservation is essential because the final framework is not only required to classify events, but also to retrieve the correct video segment. Consequently, every intermediate output is linked with a frame index and timestamp.

The video processing stage uses frame-level sampling to reduce computational cost. Processing every frame of a full-length cricket video can be expensive because broadcast videos are long and contain many visually similar frames. Sampling frames at intervals allows the framework to capture meaningful scoreboard and umpire information while maintaining practical processing time. The extracted frames are then passed to two main branches: scoreboard OCR and umpire-signal detection.

During preprocessing, frames are prepared for model inference by resizing and normalization according to the requirements of the deep learning models. For image-based model inference, frames are converted into tensors and normalized using ImageNet-style mean and standard deviation values. This is consistent with the use of pretrained CNN architectures such as EfficientNet-B0 and ResNet-18.

3.4 Scoreboard OCR Processing

Since broadcast scoreboards usually appear near the lower screen region, the scoreboard area is cropped from the lower part of the frame and passed to EasyOCR [13]. Cropping the scoreboard region reduces background noise and improves the chance that OCR focuses on relevant textual content. The scoreboard contains useful cricket context such as batsman names, bowler names, score, wickets, and over details.

The OCR output is cleaned by converting text to uppercase, replacing visually confusing characters, removing unnecessary symbols, filtering non-player terms, and extracting candidate name tokens. This cleaning step is important because raw OCR text often contains score values, separators, team abbreviations, over numbers, and unrelated broadcast text. For example, cricket scoreboards may contain text such as “FOUR”, “SIX”, “RUNS”, “OVER”, or score patterns. These terms are not player names and must be filtered before player identification.

The OCR cleaning process removes or normalizes characters that commonly appear due to recognition errors. It also filters short or irrelevant tokens and removes year-like numbers. Candidate tokens are selected based on length, alphabetic structure, and exclusion from non-name blocklists. This reduces the number of false player candidates passed to the player identification stage.

3.5 Umpire-Signal Event Detection

Key cricket events such as FOUR, SIX, and OUT are detected using umpire gestures. The event detection module follows a two-stage design. First, an EfficientNet-B0 classifier determines whether a frame contains an umpire [26]. Second, frames containing umpires are passed to a ResNet-18 pose classifier [11], which classifies the gesture into FOUR, SIX, OUT, or NO-ACTION.

The two-stage design was selected to reduce false event predictions. If pose classification is performed on all frames, the model may generate event predictions from frames that do not contain an umpire. By using a binary umpire detector first, the framework filters out irrelevant frames and passes only candidate umpire frames to the pose classifier. This makes the event detection pipeline more practical for cricket broadcast videos, where many frames show players, crowd, field, replay graphics, or scoreboard overlays rather than umpire signals.

The NO-ACTION class reduces false event detections when an umpire is visible but not signalling a major event. In this study, the OUT class represents wicket-related umpire-signal events. Confidence thresholds and frame-level grouping are applied to reduce single-frame misclassification. This is significant because an event signal may appear across several consecutive frames. If individual frames are treated independently, a single noisy prediction may create a false event. Grouping predictions around nearby timestamps helps produce cleaner event records.

3.6 Player Identification

Player identification is performed using OCR-based candidate extraction, NER-based entity recognition, and fuzzy dictionary-based player-name matching. The goal of this stage is to identify batsman 1, batsman 2, and bowler from scoreboard context. These roles are important because cricket events become more meaningful when connected to the active batsman and bowler.

OCR-based extraction first generates candidate tokens from cleaned scoreboard text. Nevertheless, OCR output can contain spelling variations, missing letters, extra characters, and partial names. Consequently, the framework uses fuzzy matching to compare extracted candidates with known player-name candidates. String similarity matching helps handle OCR errors, abbreviations, spelling variations, and partial name recognition.

The NER component improves extraction by identifying player-like entities from text. This is useful when the scoreboard contains multiple words and not all tokens are equally relevant. The NER-supported pipeline helps focus on player-related name tokens before dictionary matching. In the implemented framework, predicted labels are

used to support batsman and bowler role assignment. The final player identification output assigns three roles from scoreboard context: batsman 1, batsman 2, and bowler.

3.7 Event–Player Association

After event detection and player identification, timestamp synchronization associates detected events with the nearest available scoreboard information. For each detected event, the framework selects the closest scoreboard frame available before or after the event timestamp and attaches the corresponding batsman 1, batsman 2, and bowler predictions. The final event-player record contains the event type, timestamp, associated players, and match context.

This stage is essential because event detection and player identification are produced from different sources. The event classifier uses visual umpire frames, while player identification uses scoreboard OCR frames. These outputs must be merged correctly to create a useful retrieval record. If the event is detected but no player context is attached, the dashboard cannot answer player-specific queries. If the player is identified but no event is detected, the framework cannot generate event-based segments. Consequently, timestamp-based association connects both modalities.

Post-processing assigns ordinal labels such as *first six*, *second four*, and *last wicket*. Wicket attribution is estimated where possible using the associated bowler and scoreboard context. These labels support structured retrieval queries such as “first six”, “last wicket”, or “wickets by bowler”. The framework therefore converts frame-level predictions into higher-level cricket retrieval records.

3.8 Video Segmentation

The structured event-player records are used to generate playable video segments around detected events. Instead of requiring users to manually move through a long video, the framework stores event timestamps and prepares segments around the detected event positions. These segments allow users to directly navigate to the relevant portion of the match.

Video segmentation is based on event timestamps and associated metadata. Each generated segment is linked with the event type, timestamp, batsman information, bowler information, and available wicket attribution. This structure makes the segment searchable by multiple criteria. A user can search by event type, by player, by dismissal context, or by bowler-wicket relationship. This is more useful than simple highlight generation because the segments are not only shortened video clips but also indexed cricket events.

3.9 Structured Retrieval Methodology

The dashboard provides player-wise filtering, event-based navigation, wicket attribution display, and query-based retrieval using a rule-based filtering approach over the generated structured records. The dashboard is designed to make the processed outputs understandable and usable for cricket analysis.

User queries are parsed into predefined cricket-specific query patterns, including ordinal event queries, event-type queries, player appearance queries, player dismissal

queries, bowler-wicket queries, and invalid or unsupported queries. Examples include *first six*, *show wickets*, *Kohli batting*, *Kohli out*, and *wickets by Bumrah*. After identifying the query type, the framework searches the structured event-player records, player index, and wicket-attribution records to return the corresponding timestamped video segment.

The rule-based query approach was selected because the structured retrieval queries are domain-specific and operate on structured records generated by the pipeline. The framework does not claim unrestricted natural language question answering. Instead, it supports cricket-domain retrieval patterns that are directly connected to the implemented event-player records.

3.10 Implementation Environment

The processing pipeline implementation uses Python-based video processing and machine learning libraries. OpenCV is used for frame handling and image processing. EasyOCR is used for scoreboard text extraction. PyTorch, timm, and torchvision transformations are used for deep learning model loading and inference. The umpire detection model uses EfficientNet-B0, while the pose classification model uses ResNet-18. The player identification component uses a transformer-based NER model and fuzzy string matching. Processed outputs are organized as structured records and exposed to the dashboard for retrieval.

The structured retrieval layer connects the generated structured event-player records with user-facing retrieval features. The framework design separates video processing, event detection, player identification, record generation, and structured retrieval. This modular structure makes it easier to evaluate individual components and identify where errors occur.

4 Results and Discussion

This section evaluates the proposed approach using component-level metrics for umpire detection, event classification, OCR, player identification, event-player association, and dashboard query retrieval. The evaluation was designed to measure both individual model performance and the usefulness of the complete retrieval pipeline.

4.1 Dataset Summary

The umpire detection dataset contained 1084 training images and 270 testing images. The training set included 446 umpire and 638 non-umpire images, while the testing set included 111 umpire and 159 non-umpire images. The umpire pose dataset contained 2,000 labelled images across FOUR, SIX, OUT, and NO-ACTION classes, with 1,600 images used for training and 400 images used for testing. The scoreboard OCR dataset contained 3800 manually verified samples labelled with batsman 1, batsman 2, and bowler information. Player identification was evaluated using 800 manually verified role-label instances.

The dataset summary shows that the evaluation covers both model-level and framework-level components. The umpire detection and pose classification datasets

evaluate visual recognition. The OCR dataset evaluates scoreboard text extraction. The role-label instances evaluate player identification. The query set evaluates the structured retrieval layer. This structure provides a more complete view of framework performance than evaluating only one model.

4.2 Overall Framework Evaluation

Table 1 summarizes the major evaluation results of the proposed approach.

Table 1 Overall evaluation summary

Component	Evaluation metric	Result
Umpire detection	Accuracy / weighted F1-score	0.91 / 0.92
Event classification	Accuracy / macro F1-score	0.83 / 0.83
Scoreboard OCR	Normalized exact-match accuracy	0.9211
Player identification	OCR + NER + fuzzy matching accuracy	0.83
NER contribution	OCR + fuzzy vs. OCR + NER + fuzzy	0.79 to 0.83
Event-player association	Batsman 1 / batsman 2 / bowler accuracy	0.963 / 0.926 / 0.852
Event-player association	Full association rate	0.741
Dashboard retrieval	200-query rule-based retrieval accuracy	190/200 = 95.0%

The results show that the framework performs strongly in individual components such as umpire detection and OCR. Nevertheless, full event-player association is more difficult because it depends on multiple components working correctly together. This is expected in a multimodal pipeline because an error in OCR, player identification, event detection, or timestamp alignment can affect the final record.

4.3 Umpire Detection Performance

The EfficientNet-B0 umpire detector achieved an accuracy of 0.91 on the test set. The non-umpire class achieved precision, recall, and F1-score values of 0.95, 0.90, and 0.93, while the umpire class achieved 0.87, 0.94, and 0.90. The weighted F1-score was 0.92. The confusion matrix showed 143 correctly classified non-umpire samples and 104 correctly classified umpire samples.

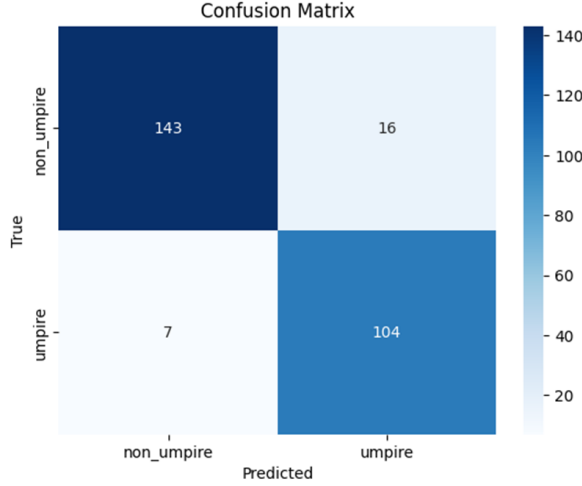
$$Umpire\ Accuracy = \frac{143 + 104}{270} = 0.9148 \approx 0.91 \quad (1)$$

The high recall for the umpire class indicates that the detector is effective at identifying frames where an umpire is visible. This is significant because missed umpire frames can prevent the event classifier from receiving relevant input. At the same time, high non-umpire performance helps reduce unnecessary pose classification on irrelevant frames. Consequently, the umpire detector supports the efficiency and reliability of the two-stage event recognition pipeline.

Some errors are still expected because cricket broadcasts contain frames where the umpire is partially visible, appears in the distance, or is occluded by players

Table 2 Classification report of umpire detection model

Class	Precision	Recall	F1-score	Support
Non-umpire	0.95	0.90	0.93	159
Umpire	0.87	0.94	0.90	111
Accuracy	—	—	0.91	270
Macro average	0.91	0.92	0.91	270
Weighted average	0.92	0.91	0.92	270

**Fig. 2** Confusion matrix of umpire detection model

or graphics. Lighting variation and camera angle changes may also affect detection. Nevertheless, the achieved performance is sufficient to support the next stage of event classification in the implemented pipeline.

4.4 Umpire Pose-Based Event Classification Performance

The ResNet-18 event classifier achieved 0.83 accuracy and 0.83 macro F1-score on the held-out pose test set. The classification task included FOUR, SIX, OUT, and NO-ACTION classes. Misclassifications mainly occurred between visually similar umpire gestures, especially when frames captured only part of the signal or when camera angle and background conditions reduced clarity.

This indicates that frame-level classification is insufficient for some visually similar umpire gestures because important motion cues are distributed across consecutive frames. For example, a single frame may not fully capture the arm motion or final gesture position. The NO-ACTION class also introduces difficulty because an umpire may be visible in a neutral pose that resembles the beginning or ending phase of a

signal. These limitations explain why the event classifier achieved lower performance than the binary umpire detector.

Table 3 Classification report of event classification model

Class	Precision	Recall	F1-score	Support
FOUR	0.91	0.84	0.87	80
NO-ACTION	0.82	0.85	0.84	99
OUT	0.83	0.78	0.80	127
SIX	0.78	0.87	0.82	94
Accuracy	—	—	0.83	400
Macro average	0.84	0.83	0.83	400
Weighted average	0.83	0.83	0.83	400

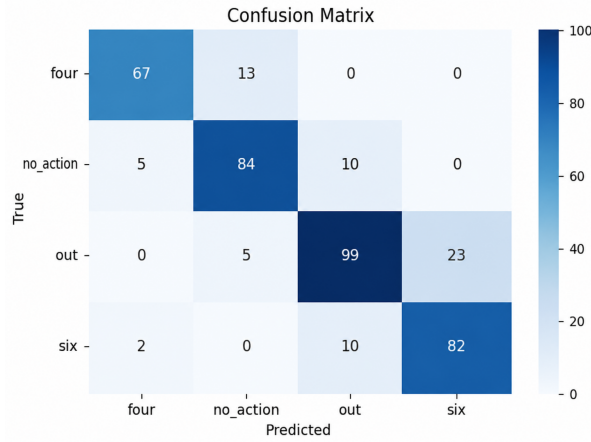


Fig. 3 Confusion matrix of event classification model

The results suggest that the ResNet-18 classifier can identify major umpire-signal events with reasonable accuracy, but temporal modelling may improve future versions. Temporal models such as LSTM, 3D CNN, or transformer-based video models may reduce these errors by learning gesture progression instead of relying only on isolated frames. Nevertheless, these temporal models were not implemented in the current framework and are identified only as future work.

4.5 Scoreboard OCR Evaluation

The OCR component was evaluated using 3800 manually verified scoreboard records. A record was considered correct when the extracted player-related scoreboard information matched the expected information after text normalization. Out of 3800 verified

OCR records, 3500 were treated as correctly extracted, resulting in a normalized exact-match accuracy of 0.9211.

$$OCR\ Accuracy = \frac{3500}{3800} = 0.9211 \quad (2)$$

Table 4 Scoreboard OCR evaluation

Metric	Result
Total OCR records	3800
Correct OCR records	3500
Normalized exact-match accuracy	0.9211

The OCR accuracy indicates that scoreboard text extraction was generally reliable after preprocessing and normalization. Nevertheless, OCR errors were mainly caused by low resolution, motion blur, compressed scoreboard text, stylized fonts, and partially visible overlays. These errors affected player-name extraction and contributed to incorrect event-player associations.

OCR errors are especially important in this framework because player identification depends heavily on scoreboard text. If a player name is missed or incorrectly read, the fuzzy matching and NER components may not receive enough reliable information. Consequently, OCR quality has a direct impact on the final player-event association result.

4.6 Player Identification and NER Contribution Evaluation

Player identification was evaluated by comparing predicted roles with manually verified labels across 800 role-label instances. The evaluation considered batsman 1, batsman 2, and bowler roles. To examine the contribution of the NER component, three configurations were compared: OCR-only extraction, OCR with fuzzy dictionary matching, and OCR with NER-supported fuzzy matching.

As shown in Table 5, OCR-only extraction achieved an overall role accuracy of 0.74. Adding fuzzy dictionary matching improved the overall accuracy to 0.79 by resolving partial names, spelling variations, and OCR token errors against the player roster. The full OCR + NER + fuzzy matching configuration achieved the highest overall role accuracy of 0.83. This indicates that NER-supported extraction improves player-role identification when scoreboard text is noisy, abbreviated, or partially recognized.

Bowler identification remained weaker than batsman identification because bowler names were less consistently visible in the scoreboard overlay. Batsman names usually appeared more prominently and consistently, which improved batsman-role identification compared with bowler-role identification. The result also shows that fuzzy matching alone improves performance, but NER provides an additional improvement by helping identify meaningful player entities from noisy OCR text.

The improvement from 0.79 to 0.83 demonstrates that the NER-supported method contributes positively to player identification. Although the increase is moderate, it

Table 5 NER contribution analysis for player identification

Method	Batsman 1	Batsman 2	Bowler	Overall
OCR only	0.80	0.82	0.61	0.74
OCR + fuzzy matching	0.83	0.86	0.68	0.79
OCR + NER + fuzzy matching	0.89	0.89	0.73	0.83

is meaningful because even small improvements in player identification can improve final event-player retrieval. In a pipeline-based framework, player identification errors can propagate into event-player association and dashboard results.

4.7 Event-Player Association Evaluation

Event-player association was evaluated by checking whether detected events were linked with the correct batsman and bowler at the corresponding timestamp. Table 6 shows that batsman 1, batsman 2, and bowler association achieved accuracies of 0.963, 0.926, and 0.852, respectively. The full event-player association rate was 0.741.

$$FAR = \frac{N_{\text{correct complete}}}{N_{\text{evaluated events}}} = 0.741 \quad (3)$$

Table 6 Event-player association performance

Metric	Score
Batsman 1 association accuracy	0.963
Batsman 2 association accuracy	0.926
Bowler association accuracy	0.852
Full event-player association rate	0.741

The lower full association rate compared with individual role accuracies indicates that complete cricket event understanding is more difficult than isolated player-role recognition. A single incorrect batsman or bowler prediction causes the complete event-player record to fail, making the final association score sensitive to OCR noise and timestamp alignment errors.

The batsman association scores were higher than the bowler association score because batsman names are generally more stable and visible on broadcast scoreboards. Bowler names may appear in smaller regions, change position depending on scoreboard layout, or be displayed less consistently. This explains why bowler association remains a challenging part of the pipeline.

4.8 Structured Retrieval Evaluation

The structured retrieval component was evaluated as a rule-based query parsing and filtering framework over structured event-player records. A manually prepared query set of 200 cricket-domain queries was created across six supported query categories: ordinal event queries, event-type queries, player appearance queries, player dismissal queries, bowler-wicket queries, and invalid/failure queries. Each query was assigned a manually verified expected output.

A query was considered correct when the framework returned the correct event, player, wicket attribution, event list, or timestamped video segment. For timestamp-based answers, the returned timestamp was compared with the manually verified reference event. Since the dashboard retrieves from generated structured records, this evaluation measures query retrieval over event-player records. This evaluation was limited to supported cricket-domain query patterns and does not represent unrestricted natural language question answering.

As shown in Table 7, the framework correctly answered 190 out of 200 supported queries, achieving an overall retrieval accuracy of 95.0%. Since each supported query returns a direct top result or a matched event list, this result also represents a top-1 retrieval accuracy of 0.950 for the evaluated query set.

Table 7 Rule-based dashboard query retrieval evaluation

Query category	No.	Supported	Correct	Accuracy
Ordinal event queries	40	40	40	100%
Event type queries	35	35	33	94.3%
Player appearance queries	35	35	35	100%
Player dismissal queries	35	35	30	85.7%
Bowler-wicket queries	35	35	32	91.4%
Invalid/failure queries	20	20	20	100%
Overall	200	200	190	95.0%

The highest results were achieved for ordinal event queries, player appearance queries, and invalid/failure queries. This is because these categories depend mainly on structured event ordering, player index matching, and rule-based rejection. Player dismissal and bowler-wicket queries achieved lower accuracy because they depend more heavily on correct event-player association and bowler attribution. If the bowler name is incorrectly extracted or the wicket context is not aligned correctly, these query types can fail.

The structured retrieval result demonstrates that structured event-player records are useful for practical cricket video navigation. Once events are correctly detected and associated with players, rule-based retrieval can return relevant video moments with high accuracy for supported query patterns.

4.9 Video Segmentation and Baseline Comparison

To evaluate the contribution of multimodal integration in video segmentation, the proposed approach was compared with single-modality baseline approaches. The purpose of this comparison was to examine how effectively each approach could identify and segment meaningful cricket video moments based on event occurrence and player-related context.

The OCR-only baseline used scoreboard OCR and player-name matching to extract textual context from video frames. Nevertheless, this approach was limited when scoreboard text was unclear, partially visible, or insufficient for identifying the actual cricket event. For example, the scoreboard may show player names and score values, but it may not clearly indicate whether the current visual moment is a four, six, or wicket. Consequently, OCR-only processing cannot reliably segment videos by visual event type.

The event-only baseline used EfficientNet-B0 for umpire presence detection and ResNet-18 for event classification without OCR-based player identification. Although this baseline could identify visual event cues, it lacked sufficient player-specific context and could not reliably associate detected events with the relevant batsman or bowler. Consequently, it was not suitable for queries such as “show boundaries by a batsman” or “show wickets by a bowler.”

In contrast, the proposed multimodal framework combines visual event detection, scoreboard-based player context, player identification, and timestamp association. This allows the framework to segment cricket videos into more meaningful player- and event-related moments. The comparison demonstrates that relying on a single modality is insufficient for accurate cricket video segmentation, while integrating visual and textual information provides stronger contextual understanding and improves the usefulness of the generated video segments.

4.10 Error Analysis

Most errors were caused by OCR noise, visually similar umpire gestures, and timestamp-based association failures. OCR errors occurred when scoreboard text was small, blurred, occluded, or displayed using stylized fonts. These errors affected player-name extraction and role assignment. Since player identification depends on OCR text, even a small OCR error can affect the final event-player record.

Event classification errors occurred between gestures such as SIX and OUT or FOUR and NO-ACTION, especially when the frame captured only part of the umpire signal. Broadcast camera angles can also reduce visibility of umpire hand positions. Since the current event classifier operates on frame-level images, it may miss motion information across consecutive frames.

Timestamp alignment errors occurred when the closest scoreboard frame did not accurately represent the player context at the exact event moment. This can happen when the scoreboard updates slightly before or after the event signal, or when the selected sampled frame is not perfectly aligned with the event. Full event-player association was more difficult than individual role prediction because one incorrect role caused the complete record to be counted as incorrect.

4.11 Limitations

The current evaluation has several limitations. Although the dataset was collected from more than 100 publicly available cricket videos with different teams, players, camera angles, lighting conditions, scoreboard layouts, and broadcast styles, generalization can still be strengthened further using larger multi-tournament benchmarks. The umpire pose dataset contained 2,000 labelled images, and the player identification evaluation used 800 manually verified role-label instances, which improves reliability compared with the earlier smaller evaluation setup. Nevertheless, additional data from more tournaments, camera styles, and scoreboard formats would further strengthen external validity.

The current event classifier focuses only on FOUR, SIX, OUT, and NO-ACTION. Other cricket events such as wides, no-balls, reviews, appeals, run-outs, and near-miss moments are not implemented. Consequently, the current framework should be understood as a major-event retrieval framework rather than a complete cricket event understanding framework. The dashboard query retrieval layer is also rule-based and supports predefined cricket-domain query patterns. It does not represent unrestricted natural language question answering.

The framework also depends on visible scoreboard information. If the scoreboard is absent, heavily blurred, or hidden by broadcast graphics, player identification becomes difficult. Similarly, if the umpire signal is not clearly visible, event classification performance may decrease. These limitations are common in broadcast video analysis and should be addressed in future work through stronger temporal modelling, improved OCR robustness, and broader evaluation.

5 Conclusion

This research presented an automated cricket video segmentation and retrieval framework for identifying player-related events in broadcast cricket videos. The framework combines scoreboard OCR, NER-supported player identification, dictionary-based name matching, and umpire-signal event detection using EfficientNet-B0 and ResNet-18. Detected events and player information are synchronized using timestamps to generate structured records for player-wise filtering, event navigation, wicket attribution, and dashboard-based query retrieval.

The results show that the proposed approach is effective for player-specific cricket video retrieval. Umpire detection achieved 0.91 accuracy and 0.92 weighted F1-score, while event classification achieved 0.83 accuracy and 0.83 macro F1-score using the 2,000-image pose dataset with 1,600 training images and 400 testing images. Scoreboard OCR achieved a normalized exact-match accuracy of 0.9211, and the full event-player association rate reached 0.741. The NER-supported player identification configuration improved overall role accuracy from 0.79 to 0.83 compared with OCR and fuzzy matching only, based on 800 manually verified role-label instances. The rule-based structured retrieval component achieved 95.0% accuracy on 200 supported cricket-domain queries.

The main contribution of this work is the integration of event detection, player identification, timestamp synchronization, and structured retrieval into one cricket-specific

workflow. The developed framework reduces manual effort by producing structured, timestamped video segments that can be directly retrieved by player identity and event type. This makes the framework useful for cricket analysis, coaching review, and content navigation.

The main limitations are restricted event classes, OCR sensitivity to broadcast quality, visible-scoreboard dependency, nearest-frame timestamp association, and the need for broader testing across more tournaments and broadcast formats. Future work will expand event categories, improve OCR robustness, introduce temporal video modelling, strengthen player-entity linking, and evaluate retrieval on larger multi-match benchmarks.

Statements and Declarations

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Ethics approval Not applicable. This study did not involve human participants, human data collected through interaction, biological material, or animal subjects.

Consent to participate Not applicable. No human participants were recruited for this study.

Consent for publication Not applicable. The manuscript does not report personally collected participant data requiring publication consent.

Data availability The datasets generated and analysed during the current study are not publicly available due to copyright restrictions associated with cricket broadcast video sources. Derived annotations may be made available from the corresponding author on reasonable request.

Materials availability Not applicable.

Code availability The source code developed for this research is not publicly available at this stage but may be made available from the corresponding author on reasonable request.

Author contribution Author contribution details are provided in the separate title page for double-blind review.

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